

Review

1. Forward Kinematics

$$x = w(q) \quad \longrightarrow \quad \frac{d x}{d t} = V(q)\dot{q}$$

Given q find x

2. Inverse Kinematics

$$q = w^{-1}(x) \quad \longrightarrow \quad \frac{d q}{d t} =$$

Given x find q

$$\dot{q} = [V^T(q)V(q)]^{-1}V^T(q)\dot{x}$$

$$\det[V^T(q)V(q)] \neq 0$$

3. Trajectory Planning

How to specify x ?

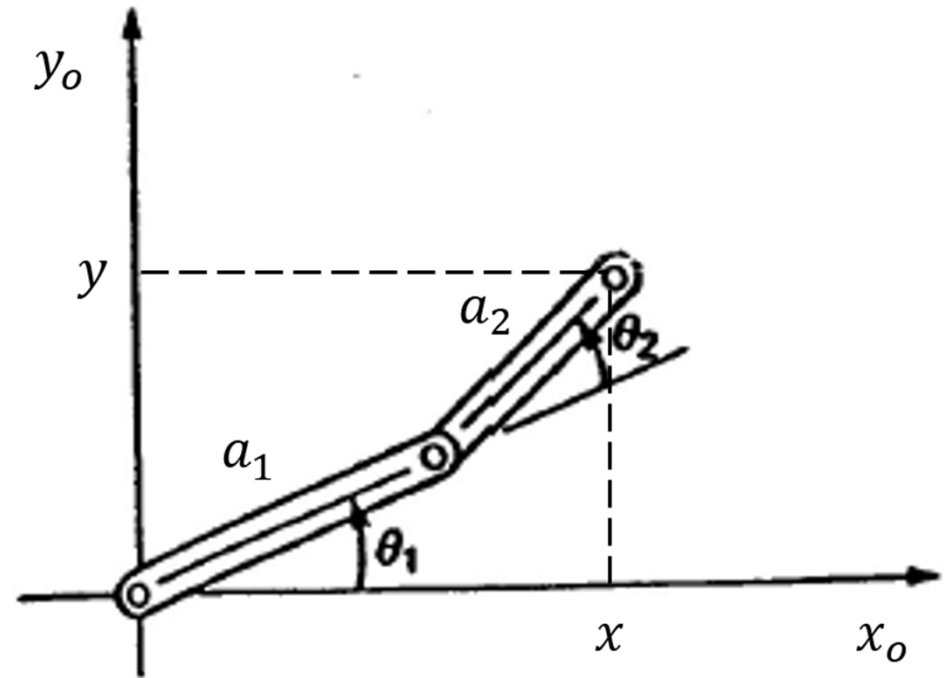
based on base coordinates obtained from DH assignments

- point to point
- trajectory $\Rightarrow$ **velocity**

Forward kinematic equations

$$x = a_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2)$$

$$y = a_1 \sin(\theta_1) + a_2 \sin(\theta_1 + \theta_2)$$



$$\begin{array}{ccccc}
 & \text{v1} & \text{v2} & \text{v3} & \text{v4} & \text{v5} \\
 V(q) = & \begin{pmatrix} X & X & X & X & 0 \\ X & X & X & X & 0 \\ 0 & X & X & X & 0 \\ X & X & X & X & X \\ X & X & X & X & X \\ 0 & X & X & X & X \end{pmatrix}
 \end{array}$$

$$\begin{aligned}
 v^1(q) &= \begin{pmatrix} -S_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) \\ C_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) \\ 0 \\ [\exp(q_5/\pi)]S_1S_{234} \\ -[\exp(q_5/\pi)]C_1S_{234} \\ 0 \end{pmatrix} \\
 v^2(q) &= \begin{pmatrix} -C_1(a_2S_2 + a_3S_{23} + a_4S_{234} + d_5C_{234}) \\ -S_1(a_2S_2 + a_3S_{23} + a_4S_{234} + d_5C_{234}) \\ -a_2C_2 - a_3C_{23} - a_4C_{234} + d_5S_{234} \\ -[\exp(q_5/\pi)]C_1C_{234} \\ -[\exp(q_5/\pi)]S_1C_{234} \\ [\exp(q_5/\pi)]S_{234} \end{pmatrix} \\
 v^3(q) &= \begin{pmatrix} -C_1(a_3S_{23} + a_4S_{234} + d_5C_{234}) \\ -S_1(a_3S_{23} + a_4S_{234} + d_5C_{234}) \\ -a_3C_{23} - a_4C_{234} + d_5S_{234} \\ -[\exp(q_5/\pi)]C_1C_{234} \\ -[\exp(q_5/\pi)]S_1C_{234} \\ [\exp(q_5/\pi)]S_{234} \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 v^4(q) &= \begin{pmatrix} -C_1(a_4S_{234} + d_5C_{234}) \\ -S_1(a_4S_{234} + d_5C_{234}) \\ -a_4C_{234} + d_5S_{234} \\ -[\exp(q_5/\pi)]C_1C_{234} \\ -[\exp(q_5/\pi)]S_1C_{234} \\ [\exp(q_5/\pi)]S_{234} \end{pmatrix} \\
 v^5(q) &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ -[\exp(q_5/\pi)]C_1S_{234}/\pi \\ -[\exp(q_5/\pi)]S_1S_{234}/\pi \\ -[\exp(q_5/\pi)]C_{234}/\pi \end{pmatrix}
 \end{aligned}$$